

9/28/22 Connor Madden

HW3

① a) $T = 327.86^\circ\text{F}$ $x = 24.98\%$ $v = 1.121 \text{ ft}^3/\text{lb}$

$h = 520.72 \text{ Btu/lb}$

b) $P = 73.359 \text{ lbf/in.}^2$ $v = 1.9959 \text{ ft}^3/\text{lb}$

$u = 327.2 \text{ Btu/lb}$

c) $T = 295.1^\circ\text{C}$ $u = 3025.7 \text{ kJ/kg}$

$h = 3348.4 \text{ kJ/kg}$

d) $P = 1.3299 \text{ bar}$ $v = .1464 \text{ m}^3/\text{kg}$

$u = 215.84 \text{ kJ/kg}$

② a) Phase 1

Sat-vapor $T_1 = 180^\circ\text{C}$

$P_1 = 10.02 \text{ bar} = 1.002 \text{ MPa}$

$v_{\text{grd}} = v_1 = v_2 = .1941 \text{ m}^3/\text{kg}$

b) Phase 2

$T_2 = 100^\circ\text{C}$

Sat Liquid-vapor

$x = \frac{v - v_f}{v_g - v_f} = \frac{(.1941 - .0010435)}{(1.673 - .0010435)}$

$x = 11.55\%$

$v = v_f + x(v_g - v_f)$

$v - v_f = x$

$\frac{v_g - v_f}{(1.673 - .0010435)}$

③ a) Phase 1

State - sat-vapor

$T_1 = 133.6^\circ\text{C}$

$P_1 = 3 \text{ bar}$

$v_1 = .6058 \text{ m}^3/\text{kg}$ $V_1 = 1.2116 \text{ m}^3$ $v_2 = 1.032 \text{ m}^3/\text{kg}$ $\therefore V_2 = 2.064 \text{ m}^3$

$m = 2$

$3 \text{ bar} = 300 \text{ kPa}$

Phase 2 b)

superheated vapor

$T_2 = 227.6^\circ\text{C}$

$P_2 = 3 \text{ bar}$

$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2} \therefore T_2 = \frac{P_2 V_2 T_1}{P_1 V_1}$

$T_2 = \frac{(2.064)(133.6)}{(1.2116)}$

$T_2 = 227.6^\circ\text{C}$

④

$W = -P(V_2 - V_1) = -300(2.064 - 1.2116) = 255.72 \text{ kJ} = W$

HW3 Cont.

④ a) Phase 1 Phase 2 b) work is done $\therefore W = -20$

$P_1 = 10 \text{ bar}$ $m = 1 \text{ kg}$ $P_2 = 2 \text{ bar}$

$T_1 = 60^\circ\text{C}$ $T_2 = 30^\circ\text{C}$

$v_1 = 0.2501 \text{ m}^3/\text{kg}$ $v_2 = 0.1185 \text{ m}^3/\text{kg}$

$U_1 = 268.35 \text{ kJ}$ $U_2 = 253.06 \text{ kJ}$

$\therefore V_1 = 0.2501 \text{ m}^3$ $Q = \Delta U - W$

$Q = (253.06 - 268.35) - (-20)$

$Q = 4.71 \text{ kJ}$

⑤ a) Phase 1 Phase 2 b) $W = 0$

liquid-vapor sat-vapor

$T_1 = 70^\circ\text{C}$ $m = 1.5 \text{ kg}$ $T_2 = 120^\circ\text{C}$

$v_1 = v_2 = 0.8919 \text{ m}^3/\text{kg}$

$P_1 = 0.3119 \text{ bar} = 31.2 \text{ kPa} = P_2$

$u_1 = 781.17 \text{ kJ/kg}$ $u_2 = 252.93 \text{ kJ/kg}$

$Q = \Delta U = \Delta u \cdot m$

$Q = (u_2 - u_1)(m)$

$Q = (252.93 - 781.17)(1.50)$

$Q = -764.625 \text{ kJ}$

$u = u_f + x(u_g - u_f)$

$252.93 = 781.17 + x(252.93 - 781.17)$

$x = 0.1767$

$v_1 = 677.585$ $v_g - v_f = 5.042 - 0.001023$

⑥ a) $P_1 = 10 \text{ bar}$ $x = 60\%$ $m_g = 25 \text{ kg}$

$\therefore M = m_g + m_f = 34.375 \text{ kg}$

$V = v \cdot m = (v_f + x(v_g - v_f))(m)$

$V = (0.008352 + 0.6(0.0236 - 0.008352))(34.375) = 0.498 \text{ m}^3$

$m_g = 9.375 \text{ kg}$

b) $\frac{v_g \cdot m \cdot x}{V} = \frac{0.0236 \cdot 34.375 \cdot 0.6}{0.498} = 97.695\%$

$\frac{0.48675}{0.49823}$

⑦ a) $R = \bar{R}/M = \frac{8.314 \text{ (kJ)}}{4.003 \text{ (kmol} \cdot \text{K)}} \cdot \left(\frac{\text{kmol}}{\text{kg}}\right) = 2.077 \text{ J/kg} \cdot \text{K} = R$

b) $\frac{Pv}{T} = R \therefore v = \frac{RT}{P} \therefore V = \frac{nRT}{P} = (0.25) \left(\frac{2.077 \text{ J/kg} \cdot \text{K}}{1000 \text{ J/m}^3 \cdot \text{K}}\right) (300 \text{ K}) = 0.0096 \text{ m}^3$

c) $\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2} \therefore V_2 = \frac{P_1 V_1 T_2}{P_2 T_1} = \frac{100000 \text{ N/m}^2 \cdot 0.0096 \text{ m}^3 \cdot 300 \text{ K}}{150000 \text{ N/m}^2 \cdot 300 \text{ K}} = V_2 = 959 \text{ cm}^3$

HW3 Cont.

(8) a) Phase 1

$$T_1 = 298^\circ\text{K}$$

$$P_1 = 275 \text{ kPa}$$

$$V_1 = V_2 = .05 \text{ m}^3$$

Phase 2

$$T_2 = 530^\circ\text{K}$$

$$P_2 =$$

$$V_2 = V_1 = .05 \text{ m}^3$$

$$W = 2.75 \text{ kJ} = \Delta U$$

$$Q = -2.75 \text{ kJ}$$

$$PV = mRT \rightarrow \frac{PV}{RT} = m = \frac{(275 \text{ kPa})(.05 \text{ m}^3)}{(.1889 \frac{\text{kJ}}{\text{kg} \cdot \text{K}})(298 \text{ K})} = .244 \text{ kg} = m$$

$$R = \frac{\bar{R}}{M} = \frac{8.314 \text{ kJ/mol} \cdot \text{K}}{44.01 \text{ kg/kmol}} = .1889 \frac{\text{kJ}}{\text{kg} \cdot \text{K}}$$

b) $W = \Delta U + 2.75 \text{ kJ}$

$$\Delta U = U_2 - U_1$$

$$\Delta U = (\bar{U}_2 - \bar{U}_1)n$$

$$U_1 = n \times \bar{U}_1$$

$$U_2 = n \times \bar{U}_2$$

$$W = 2.75 \text{ kJ} + (\bar{U}_2 - \bar{U}_1) \frac{m}{M}$$

$$\begin{aligned} \bar{U}_1 &= 6885 \text{ kJ/kmol} \\ \bar{U}_2 &= 14622 \text{ kJ/kmol} \end{aligned} \quad \begin{aligned} n &= \frac{m}{M} \\ m &= .244 \text{ kg} \\ M &= 44.01 \text{ kg/kmol} \end{aligned}$$

$$W = 2.75 + (14622 - 6885) \left(\frac{.244}{44.01} \right) = 2.75 \text{ kJ} + 42.94 \text{ kJ}$$

$$W = 45.69 \text{ kJ}$$